

Signed-first Cartan partition-refinement obstruction and rational ridge-deficit boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-27 · this version issued 2026-07-27 · v1.0

1 Purpose and scope

R-097 reduced the complete heat-lifted current problem to one terminal posterior bracket

$$\mathfrak{C}_{\text{post}}(B; R) = \sum_i \int \left\{ \mathbb{E}_{\mathcal{F}}[B : (G_i^{\otimes 2} - \Gamma_i)] - q_i^T (\overline{B} + 2R)^{-1} q_i \right\} dx, \quad (1.1)$$

where $\overline{B} = \mathbb{E}_{\mathcal{F}} B$ and $q_i = \mathbb{E}_{\mathcal{F}}(BG_i)$. It also left the standalone Cartan one-use ledger R-085 (4.11) and the rational shifted-Hessian form R-085 (6.5) open. This note tests those three surfaces together, with the complete signed endpoint retained.

Five exact conclusions result.

First, the posterior bracket is superadditive under a matching split of the matrix payment. This makes a rowwise lower estimate legal, although it may discard helpful cross-row compensation and gives no positivity.

Second, the production rational rank-one Gram has an exact conditional ridge-regression form. It identifies the missing analytic estimate without revealing the coefficient or differentiating an adapted selector.

Third, R-085 (6.5) is recovered exactly from the R-097 terminal identity only after the five unshifted families, the low endpoint, and the Cameron–Martin matrix payment are restored once. A lower estimate for the unshifted families is not the upper estimate needed by this recovery.

Fourth, a genuine production Cartan slice gives an anisotropic reverse partition of one source-block visit. Each oppositely oriented subvisit has a nonzero arbitrarily far harmonic whose squared atom is order A^2 , while the mixed energy–terminal-sextic budget is only order A . The two signed atoms cancel exactly before squaring. Therefore no per-subvisit nonnegative R-085/R-088 atomization can be stable under this partition refinement while retaining only the mixed payload. This does not address two visits carrying distinct temporal roots or heat operators, and it does not touch the regular one-shot theorem.

Fifth, independent-root resampling gives a derivative-free successor. The production Gram obeys a global finite-difference inequality and the bare future-control kernel is

$$\sum_{j < k} 2^j 2^{-4k} < 2^{-3k}. \quad (1.2)$$

The remaining work is to prove the spatial mixed-norm estimate with the Schur mean, sextic multiplier, trace, heat, low endpoint, and R-063 forest each retained once.

These statements do not prove the production posterior lower form, the standalone Cartan one-shot ledger, rational (6.5), complete H_N , REG, arbitrary progressive assembly, $\text{OVERLAP}_{\text{src}}$, the $q = 10/9$ Nelson estimate, a measure theorem, or Sector-A closure. Tier stays T4.

2 Posterior bracket and payment-split superadditivity

Fix one conditional fibre and suppress i, x . For a random positive semidefinite matrix B , a deterministic $R \succ 0$, and a square-integrable vector G , assume that every displayed expectation below is finite (this is automatic at finite production cutoff), and put

$$\mathcal{C}(B; R) = \mathbb{E}_{\mathcal{F}}[B : (G^{\otimes 2} - \Gamma)] - q_B^T (\bar{B} + 2R)^{-1} q_B, \quad q_B = \mathbb{E}_{\mathcal{F}}(BG). \quad (2.1)$$

This is the shorter form of the R-097 identity $J_B + \mathbb{E}_{\mathcal{F}}[B : (V_B - \Gamma)]$.

Theorem 2.1 (payment-split posterior superadditivity)

Let $B = \sum_{a=1}^r B_a$ and $R = \sum_{a=1}^r R_a$, with $B_a \succeq 0$ and $R_a \succ 0$. Then

$$\boxed{\mathcal{C}(B; R) \geq \sum_{a=1}^r \mathcal{C}(B_a; R_a).} \quad (2.2)$$

Proof. Set $A_a = \mathbb{E}_{\mathcal{F}} B_a + 2R_a$, $q_a = \mathbb{E}_{\mathcal{F}}(B_a G)$, $A = \sum_a A_a$, and $q = \sum_a q_a$. With $y = A^{-1}q$,

$$\begin{aligned} \sum_a q_a^T A_a^{-1} q_a - q^T A^{-1} q &= \sum_a \left\{ q_a^T A_a^{-1} q_a - 2y^T q_a + y^T A_a y \right\} \\ &= \sum_a \|A_a^{-1/2} q_a - A_a^{1/2} y\|^2 \geq 0. \end{aligned} \quad (2.3)$$

The Wick part of (2.1) is linear in B , so (2.3) proves (2.2). $\square$

The same calculation shows that increasing R increases $\mathcal{C}$. This is not a free improvement in the original action: the terminal Schur identity simultaneously spends the actual term $\int c^T R c$. Equation (2.2) permits a lower proof by rows, but the full bracket can be strictly larger because it keeps cross-row regression compensation.

3 The rational row is an exact ridge deficit

For the production rational row write

$$B_R = b \otimes b, \quad b = 2\sqrt{c_1} g(Z_*), \quad s_i = b^T G_i. \quad (3.1)$$

Condition on a sigma-field making the terminal control and R measurable, assume the displayed conditional moments are finite, and put

$$K = \mathbb{E}_{\mathcal{F}}(bb^T), \quad q_i = \mathbb{E}_{\mathcal{F}}(bs_i), \quad A_R = K + 2R. \quad (3.2)$$

Theorem 3.1 (conditional rational ridge normal form)

For every derivative direction i ,

$$\boxed{\begin{aligned} \mathcal{C}_i(B_R; R) &= \mathbb{E}_{\mathcal{F}}[s_i^2 - b^T \Gamma_i b] - q_i^T A_R^{-1} q_i \\ &= \inf_{h \in \mathbb{R}^6} \left\{ \mathbb{E}_{\mathcal{F}}(s_i - b^T h)^2 + 2h^T R h - \mathbb{E}_{\mathcal{F}}(b^T \Gamma_i b) \right\}. \end{aligned}} \quad (3.3)$$

Proof. The first equality is (2.1) with $B = b \otimes b$. Expanding the expression inside the infimum gives

$$\mathbb{E}_{\mathcal{F}} s_i^2 - 2h^T q_i + h^T (K + 2R) h - \mathbb{E}_{\mathcal{F}}(b^T \Gamma_i b). \quad (3.4)$$

Its unique minimizer is $h = A_R^{-1} q_i$, which gives (3.3). $\square$

Thus the missing sign is a weighted conditional ridge deficit: how much of the derivative carrier $s_i = b^T G_i$ can be predicted by the random ridge $b^T h$, compared with the covariance-normal trace. Positivity of B_R does not decide that deficit.

4 Exact recovery boundary for R-085 (6.5)

Use the R-085 notation

$$F_{6.5} = \mathcal{H}_R + \frac{1}{2} \sum_i \int c_i^T B_1 c_i dx, \quad W_0 = \frac{1}{2} \sum_i \int B_0 : (G_i^{\otimes 2} - \Gamma_i) dx, \quad (4.1)$$

and let $\mathcal{U}_R$ be the sum of the five unshifted families. The exact Taylor identity is

$$\Delta W_R = \mathcal{U}_R + F_{6.5}. \quad (4.2)$$

Theorem 4.1 (standalone recovery from the terminal bracket)

With $\overline{\mathcal{A}} = \mathbb{E}_{\mathcal{F}} B_1 + 2R$ and $q_i = \mathbb{E}_{\mathcal{F}}(B_1 G_i)$,

$$\boxed{\begin{aligned} \mathbb{E}_{\mathcal{F}} F_{6.5} &= \frac{1}{2} \sum_i \int \|c_i + \overline{\mathcal{A}}^{-1} q_i\|_{\overline{\mathcal{A}}}^2 dx + \frac{1}{2} \mathfrak{C}_{\text{post}}(B_R; R) \\ &\quad - \sum_i \int c_i^T R c_i dx - \mathbb{E}_{\mathcal{F}} W_0 - \mathbb{E}_{\mathcal{F}} \mathcal{U}_R. \end{aligned}} \quad (4.3)$$

Proof. The R-097 terminal Schur identity says

$$\mathbb{E}_{\mathcal{F}} W_R(A^*) + \sum_i \int c_i^T R c_i = \frac{1}{2} \sum_i \int \|c_i + \overline{\mathcal{A}}^{-1} q_i\|_{\overline{\mathcal{A}}}^2 + \frac{1}{2} \mathfrak{C}_{\text{post}}. \quad (4.4)$$

Insert $W_R(A^*) = W_0 + \mathcal{U}_R + F_{6.5}$ from (4.2) and rearrange. $\square$

Every owner in (4.3) occurs once. In particular, a lower bound on $F_{6.5}$ requires an *upper* form bound on $\mathcal{U}_R$, because it appears with a minus sign. R-085 records a lower bound for its five-family sum. Its familywise absolute estimates suggest a two-sided statement, but that upper statement is not a current accepted authority and is not inferred here. R-097 terminalizes $\mathcal{U}_R + F_{6.5}$, not $F_{6.5}$ alone.

5 Exact rational production-fibre sign

On the pure-doublet ray the rational coefficient is

$$B_R(u) = 4c_1 \left(u - \frac{5}{9} \frac{u^3}{u^2 + e} \right)^2, \quad c_1 = \frac{243}{8000P}. \quad (5.1)$$

It is strictly increasing for $u > 0$: after writing $u = t\sqrt{e}$, the derivative of the dimensionless squared factor is

$$\frac{2t(4t^2 + 9)(2t^2 - 3t + 3)(2t^2 + 3t + 3)}{81(t^2 + 1)^3} > 0 \quad (t > 0).$$

At the two audit points $u_- = \frac{1}{2}\sqrt{e}$ and $u_+ = \frac{3}{2}\sqrt{e}$,

$$B_- = \frac{3e}{125P}, \quad B_+ = \frac{2187e}{21125P}, \quad \Delta = B_+ - B_- = \frac{1680e}{21125P}. \quad (5.2)$$

Use the positive-branch inverse to construct the bounded even selector for which

$$B_R(u(G^2)) = B_- + \frac{\Delta}{1 + G^2}. \quad (5.3)$$

Let $G = 0$ with probability $1/2$ and $G = \pm\sqrt{2}$ with probability $1/4$ each. Then $\Gamma = 1$ and symmetry gives $q = r = 0$ for every $R > 0$. Hence

$$\boxed{\mathcal{C}(B_R; R) = \mathbb{E}[B_R(u(G^2))(G^2 - 1)] = -\frac{\Delta}{3} = -\frac{560e}{21125P} < 0.} \quad (5.4)$$

This is an exact production-coefficient fibre obstruction to automatic ridge positivity. It has no torus control map, spatial derivative $c = Da$, heat/low packet, or Cameron–Martin realization, and therefore is not a counterexample to a paid production lower form.

The complete four-row pure-doublet coefficient from R-093 gives an even stronger already accepted sign test. The R-097 posterior identity shows that it is literally a negative $\mathfrak{C}_{\text{post}}$ example with $J_B = r = 0$, not merely a negative fragment of the earlier perspective formula. This transfer adds no torus counterexample.

6 An anisotropic Cartan partition-refinement obstruction

Fix the production Cartan generator

$$S = \text{diag}(1, -1) \quad (6.1)$$

inside one active real doublet. Let $n_j \in \mathbb{Z} \setminus \{0\}$ be an admissible torus carrier whose physical frequency $N_j = 2\pi|n_j|/L$ lies in shell j , let $0 < \epsilon < 1$, and set

$$f_\epsilon(\theta) = (\sin \theta, \epsilon \cos \theta), \quad \theta = \frac{2\pi n_j x_1}{L}. \quad (6.2)$$

Since $|f_\epsilon| \geq \epsilon$, this path remains uniformly away from the cone tip at every fixed positive floor. Refine one deterministic source- block secant into two oppositely oriented subvisits, both evaluated with the same fixed target heat and the same root derivative, with

$$a_+ = Af_\epsilon, \quad a_- = -Af_\epsilon, \quad (6.3)$$

and use the fresh Cameron–Martin direction $v = \sqrt{\lambda_{n_j}} f_\epsilon$.

For the production heat-lifted Cartan one-form $\mathcal{C}_\Phi(z) = \Phi(z)^T Dz$, define

$$\begin{aligned} T_A^+ &= \mathbb{D}_j[\mathcal{C}_\Phi(w + Af_\epsilon) - \mathcal{C}_\Phi(w)][v], \\ T_A^- &= \mathbb{D}_j[\mathcal{C}_\Phi(w) - \mathcal{C}_\Phi(w + Af_\epsilon)][v]. \end{aligned} \quad (6.4)$$

Endpoint reversal gives the pathwise identity

$$\boxed{T_A^- = -T_A^+}. \quad (6.5)$$

Put

$$d_\epsilon(\theta) = \sin^2 \theta + \epsilon^2 \cos^2 \theta, \quad q_\epsilon(\theta) = \frac{\sin^2 \theta - \epsilon^2 \cos^2 \theta}{d_\epsilon(\theta)}. \quad (6.6)$$

Theorem 6.1 (nonzero far harmonic and signed cancellation)

At fixed cutoff, fixed future heat, and fixed positive floor,

$$\frac{T_A^+}{AN_j \sqrt{\lambda_{n_j}}} \longrightarrow \tau_\epsilon(\theta) := q_\epsilon(\theta) d'_\epsilon(\theta) \quad \text{in } L^2(\Omega \times \mathbb{T}^3) \quad (6.7)$$

as $A \rightarrow \infty$. If

$$\rho = \frac{1 - \epsilon}{1 + \epsilon}, \quad \phi = 2\theta,$$

then

$$q_\epsilon = \rho - \frac{4\epsilon}{1 - \epsilon^2} \sum_{n \geq 1} \rho^n \cos(n\phi), \quad (6.8)$$

and, for every $k \geq 2$, the sine-series coefficient is

$$\boxed{[\sin(k\phi)]\tau_\epsilon = -2\epsilon(1 - \rho^2)\rho^{k-1} \neq 0.} \quad (6.9)$$

Proof. Write the positive production floor as $\eta > 0$ and $n_\epsilon = \sin^2 \theta - \epsilon^2 \cos^2 \theta$, so the actual scalar coefficient on the radial slice is $A^2 n_\epsilon / (A^2 d_\epsilon + \eta)$. At zero base and zero fixed heat displacement, direct radial differentiation of the complete one-form gives the finite- A identity

$$\frac{T_A^+}{AN_j \sqrt{\lambda_{n_j}}} = \tau_\epsilon(\theta) \left\{ 1 - \frac{\eta^2}{(A^2 d_\epsilon(\theta) + \eta)^2} \right\}. \quad (6.7a)$$

Thus the floor correction is nonzero at finite A —the floorless homogeneity shortcut would be false—but it is uniformly $O(A^{-4})$ because $d_\epsilon \geq \epsilon^2$. In the leading floorless limit the two chain-rule lines $D\Phi[v]^T Dz$ and $\Phi^T Dv$ each contribute half of $AN_j \sqrt{\lambda_{n_j}} q_\epsilon d'_\epsilon$. A fixed base and fixed heat displacement are lower order on $|Af_\epsilon| \geq A\epsilon$; the global production derivative bounds and the finite-cutoff moment bounds give L^2 domination. This proves (6.7).

Writing d_ϵ as an affine function of $\cos \phi$ and using the Poisson-kernel series gives (6.8). Multiplication by $d'_\epsilon = (1 - \epsilon^2) \sin \phi$ and the elementary product-to-sum identity gives (6.9) for $k \geq 2$. $\square$

For the audit value $\epsilon = 1/2$, $\rho = 1/3$ and the corrected closed form is

$$[\sin(2k\theta)]\tau_{1/2} = -\frac{8}{3^{k+1}}, \quad k \geq 2. \quad (6.10)$$

In particular the $k = 2, 3$ coefficients are $-8/27$ and $-8/81$. The power $k + 1$ in (6.10) is independently recomputed rather than inferred from a decimal fit.

Fix a gap $C \geq 5$. Choose k_C so that the physical carrier $2k_C N_j$ lies in a nonzero smooth Littlewood–Paley block Π_m with $m \geq j + C$. The leading term in (6.7) is independent of the fresh probability root, so its OU image is unchanged. Since $\int_0^\infty e^{-2t} dt = 1/2$, (6.7)–(6.9) imply that for all sufficiently large finite A ,

$$\int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(j)} T_A^\pm\|_{\text{HS}}^2 dt \geq c_{C,\epsilon,n_j} A^2, \quad (6.11)$$

where $c_{C,\epsilon,n_j} > 0$. The one-direction limiting coefficient before the fixed LP multiplier is

$$N_j^2 \lambda_{n_j} \epsilon^2 (1 - \rho^2)^2 \rho^{2(k_C-1)}. \quad (6.12)$$

Corollary 6.2 (nonnegative per-subvisit atomization fails)

Suppose an extension of R-085 (4.10)–(4.11), or of the corrected R-088 direct-root analogue, assigns nonnegative atom ledgers q_+, q_- to these two subvisits, is stable under refinement of one source-block secant, and has a constant uniform in A . Equation (6.11) forces

$$q_+ + q_- \geq cA^2. \quad (6.13)$$

The complete loop has terminal shift zero. With the fixed positive source- cost coefficient

$$\kappa_{n_j,\epsilon} := \|K_{n_j}^{-1} f_\epsilon\|_{\mathcal{H}}^2,$$

its cost is $X = 2\kappa_{n_j,\epsilon} A^2$, while its terminal sextic Y is independent of A . Consequently

$$1 + \mathbb{E}[X^{1/2} Y^{1/2}] = O(1 + A), \quad (6.14)$$

contradicting a uniform bound of $q_+ + q_-$ by (6.14).

There is no contradiction after signed grouping: (6.5) gives $T_A^+ + T_A^- = 0$ before any square. Nor is there a contradiction if one allows a once-only pure X payment, since (6.13) is then critical rather than supercritical. Distinct temporal visits need not share the target heat or derivative root, so (6.5) is not asserted for them. The result is exactly a no-go for refinement-stable nonnegative per-subvisit atomization; R-085's regular physical-shell one-shot target and R-092's regular H_C theorem remain intact.

7 Derivative-free causal finite differences

The exact production frame bounds are

$$\|M_A(z)\|_F \leq \sqrt{20}|z|, \quad \|M_A(z) - M_A(w)\|_F \leq \sqrt{68}|z - w|. \quad (7.1)$$

For

$$B_{\text{fr}}(z) = \sum_{A=1}^3 M_A(z) Q_{\Pi} M_A(z)^T, \quad (7.2)$$

polarization gives the global secant estimate

$$\|B_{\text{fr}}(z) - B_{\text{fr}}(w)\|_F \leq C_B(|z| + |w|)|z - w|, \quad C_B = 3\|Q_{\Pi}\|_{\text{op}}\sqrt{20 \cdot 68}. \quad (7.3)$$

Let $h_k^{(j)}$ be obtained from a strict-past source h_k by independently resampling root $j < k$. Here Var_j is the Efron–Stein conditional variance with all roots other than j held fixed; it is not the Doob variance conditional only on $\mathcal{F}_{j-1}$. Conditional variance gives

$$\mathbb{E}\|h_k - h_k^{(j)}\|^2 = 2\mathbb{E}\text{Var}_j(h_k) \leq 2\mathbb{E}\|h_k\|^2. \quad (7.4)$$

If $a_k = K_k h_k$ and $\|K_k\|_{L^2 \rightarrow L^2} \leq \kappa_K 2^{-2k}$, then

$$\mathbb{E}\|a_k - a_k^{(j)}\|_2^2 \leq 2\kappa_K^2 2^{-4k} \mathbb{E}\|h_k\|^2. \quad (7.5)$$

For $k > j_0$, after the displayed resampling prefactor $2\kappa_K^2$ is kept outside, the derivative-root covariance weights have the exact scalar Hardy mass

$$\sum_{j=j_0}^{k-1} 2^j 2^{-4k} = (2^k - 2^{j_0}) 2^{-4k} < 2^{-3k}. \quad (7.6)$$

Equations (7.3)–(7.6) require no Malliavin derivative of h_k . They provide a candidate successor to per-subvisit nonnegative atomization, but are not yet an ANOVA decomposition or a once-only covariance ledger: summing raw root influences counts a Hoeffding component once for every root in its support. They also do not control the product multiplier $|Z| + |Z^{(j)}|$, the conditional Schur mean $q^T(\overline{B} + 2R)^{-1}q$, or the spatial low-Hermite/Wick pairing. The next theorem must combine those objects before using (7.6).

8 Proof and failure map

Route	Verdict	Exact boundary / successor
Posterior row split	proved	Superadditive only with a matching payment split; no row positivity.
Rational rank-one packet	proved reduction	Exact ridge deficit (3.3); spatial paid bound remains open.
R-085 (6.5) from R-097	repaired reduction	Requires the low endpoint, CM cost, and an upper form for the five unshifted families.
Rational automatic sign	failed	Exact production fibre gives (5.4); not a torus/control counterexample.
Per-subvisit nonnegative Cartan q	failed	Opposite refinements cost A^2 separately but cancel signed; mixed budget is only A .
Signed-first grouping	required	Reverse-refinement cancellation and R-097 terminalization agree exactly.
Causal resampling	advanced	Frame secant and bare 2^{-3k} scalar mass proved; once-only decomposition, mixed spatial norm, and owner assembly open.

The dependency order is now

$$\begin{array}{l}
 \text{complete R-077 cancellation} \longrightarrow \text{R-097 Gram terminalization} \\
 \longrightarrow \text{signed grouped posterior/resampling form} \longrightarrow H_N \text{ and lossless temporal assembly} \\
 \longrightarrow \text{OVERLAP}_{\text{src}} \equiv \text{the } q = 10/9 \text{ Nelson objective.}
 \end{array} \tag{8.1}$$

The standalone one-shot Cartan and rational statements remain useful sufficient modules, but nonnegative revisit atomization is no longer a valid route to the last line.

9 Devil's-advocate self-test

- Objection: posterior superadditivity proves positivity. DISMISSED.** It only compares the full bracket with the sum of row brackets. The rational production fibre (5.4) is negative.
- Objection: a larger matrix payment makes the bracket positive for free. DISMISSED.** The bracket is monotone in R , but the original action simultaneously subtracts the once-only cost $\int c^T R c$.
- Objection: R-097 already proves R-085 (6.5). UPHELD as false.** It terminalizes $\mathcal{U}_R + F_{6.5}$. Equation (4.3) retains $-\mathbb{E}\mathcal{U}_R$, so a suitable upper form is still required.
- Objection: the rational sign fixture disproves the paid torus packet. DISMISSED.** It has no spatial control realization, low endpoint, heat packet, forest, or Cameron–Martin cost.
- Objection: reverse subvisits contradict the exact full action. DISMISSED.** The two atoms cancel identically in (6.5). The contradiction applies only after squaring and assigning nonnegative ledgers to the visits separately.
- Objection: the far harmonic disappears at a positive floor or after target heat. DISMISSED.** The anisotropic path stays at distance at least $A\epsilon$ from the tip, and (6.7) is an L^2 limit with fixed heat and floor. Every fixed gap has a nonzero coefficient (6.9).

7. **Objection: at $\epsilon = 1/2$ the coefficient is $-8/3^{k+2}$. DISMISSED BY INDEPENDENT FACTOR AUDIT.** Substitution in (6.9) gives $-8/3^{k+1}$; direct quadrature gives $-8/27$ at $k = 2$ and $-8/81$ at $k = 3$.
8. **Objection: the resampling kernel closes the posterior form. UPHELD as premature.** Equation (7.6) is the bare covariance/control kernel. The spatial multiplier, low-Hermite pairing, Schur mean, sextic growth, trace, heat, low endpoint, and forest still require one integrated estimate.
9. **Objection: the new no-go withdraws R-092 regular H_C . DISMISSED.** The fixture refines one source-block secant into opposite subvisits with the same heat/root direction. R-092 is explicitly a mutually orthogonal no-revisit one-shot theorem.

External review should focus on the radial-chain factor in (6.7), the Fourier normalization in (6.9)–(6.10), the payment split in (2.2), and the minus sign on $\mathcal{U}_R$ in (4.3).

10 Authorities and executable verification

The load-bearing repository authorities are R-066 for the production frame bounds, R-079 for exact endpoint ownership, R-085 for the Cartan and rational targets, R-088 for the corrected direct-root ledger, R-092 for regular Cartan closure, R-093 for the complete production sign slice, R-096 for cancellation-first ordering, and R-097 for terminal Gram/posterior form. A targeted search of the external Contents archive found no theorem matching the production posterior/ridge or reverse-visit estimate.

The primary executable uses exact rational arithmetic, high-resolution Fourier quadrature, random positive matrices, and deliberate sign/factor mutations. The independent executable uses only the Python standard library and separately recomputes the fractions, Fourier coefficients, scalar and matrix ridge identities, superadditivity, reverse-loop exponents, and Hardy kernel. The integrated verifier pins all authorities, sources, the proof PDF, exploration records, negative-result entry, public surfaces, and assertion counts.

Run from the repository root:

```
python codes/foundations/
a13_classii_signed_first_cartan_rational_ridge_boundary_verify.py
```

Result footer

Result ID: A13-CLASSII-SIGNED-FIRST-CARTAN-RATIONAL-RIDGE-BOUNDARY

Ledger: R-098.

Title: Signed-first Cartan partition-refinement obstruction and rational ridge-deficit boundary.

Precise statement:

The terminal posterior bracket is superadditive under a matching row/payment split. The rational rank-one row is exactly a conditional ridge deficit, and standalone R-085 (6.5) has the once-only recovery identity (4.3). An exact rational production fibre remains negative. A production anisotropic Cartan reverse partition has nonzero far harmonics of squared size A^2 at each subvisit, while the mixed $X^{(1/2)}Y^{(1/2)}$ budget is $O(A)$; the signed atoms cancel exactly. Production frame resampling has the bare derivative-free Hardy kernel below $2^{(-3k)}$.

Scope: Finite cutoff and fixed positive floor. Posterior identities are conditional finite-dimensional forms. The Cartan obstruction is a deterministic opposite refinement of one source-block secant, sharing its target heat and root derivative, for every fixed gap. It does not assert cancellation for distinct temporal roots/heats and does not apply to the regular one-shot class. The rational sign

fixture is not an admissible torus control.
 Dependencies:
 R-066; R-079; R-085; R-088; R-092; R-093; R-096; R-097.
 Evidence grade:
 ANALYTIC, EXACT, EXECUTED, BOUNDARY.
 Reproduction command:
 python codes/foundations/a13_classii_signed_first_cartan_
 rational_ridge_boundary_verify.py
 Expected output:
 Manifest-pinned primary, standard-library independent,
 integrated, and aggregate PASS; exit zero.
 Negative result:
 NG-2026-07-27-A13-NONNEGATIVE-PER-SUBVISIT-CARTAN-ATOMIZATION.
 Falsification gate:
 Failure of posterior superadditivity, the rational ridge or
 recovery identity, any exact fraction, the Cartan Fourier
 coefficient/reverse cancellation/scaling, resampling/Hardy
 kernel, authority, hash, PDF, exploration, registry, public
 surface, count, or T4 scope check.
 Tier before / after:
 T4 / T4; no promotion.
 No-overclaim statement:
 No production posterior lower form, standalone Cartan (4.11),
 rational (6.5), complete H_N , REG, arbitrary progressive/revisit
 assembly, OVERLAP_src, Nelson, cutoff/floor removal, measure,
 T5--T7, or Sector-A closure is proved.
 Next required action:
 Prove a signed grouped causal-resampling posterior form in spatial
 mixed norms, retaining the Schur mean, sextic multiplier, trace,
 heat, low endpoint, CM payment, and R-063 forest once.